

Practice set 2

Exponentials and logarithms

Write in the spaces

- (c) Jessica is investigating a compounding investment. She wants to know how long it would take for an investment of \$1000 to double in value to \$2000. She forms the following equation:

$$2000 = 1000 \left(1 + \frac{R}{100} \right)^D$$

where R is the rate of return on the investment, as a percentage, and

D is the time that the investment would take to double in value, in years.

- (i) If an investment takes 11 years to double in value, what is its rate of return?

- (ii) By making D the subject of the expression: $2000 = 1000 \left(1 + \frac{R}{100} \right)^D$,

show that $D = \frac{\log(2)}{\log(1 + \frac{R}{100})}$

In her research, Jessica comes across a simple but **approximate** rule for calculating D , the time that the investment would take to double in value. It is commonly called the ‘Rule of 72’, and it states that:

$$D = \frac{72}{R}$$

Jessica wonders how close the values of D from the ‘Rule of 72’ are to those calculated using the actual expression, which is:

$$D = \frac{\log(2)}{\log(1 + \frac{R}{100})}$$

- (iii) Show clearly that the value of R for which the ‘Rule of 72’ exactly calculates D , is the solution to the equation:

$$2^R - \left(1 + \frac{R}{100}\right)^{72} = 0$$

You do not need to solve this equation.

(ii) Solve the equation $2^x = 2022$.

(b) Simplify the following expression fully, writing your answer as a single logarithm.

$$\log(3a) + 2\log\left(\frac{a}{6}\right)$$

(c) Consider the equation $\log_2(x - a) - \log_2(x + a) = c$, where a and c are constants.

(i) Show that when x is made the subject of this equation, $x = a \frac{1+2^c}{1-2^c}$.

Ensure that you use correct mathematical statements in your reasoning.

Question Three continues
on the following page

- Explaining your reasoning clearly, describe which values of a and of c will make the equation possible to solve.

You may find it useful to recall that, when x is made the subject of this equation, $x = a \frac{1+2^c}{1-2^c}$.